

AI BlogNest API

AI-Powered Blog Management and Content Generation REST API Platform

PHASE 4: PROJECT PLANNING

Role	Team Member
Team Leader	Praveen Kumar M
Team Member 1	Prabu R
Team Member 2	Muralidharan S
Team Member 3	Yadhava Moorthy Y

Submitted as a phase-wise project submission.

1. Development Plan

Stage	Planned Work Description
Stage 1: Architecture & Setup	Project initialization, directory scaffolding, Express server setup (<code>server.js</code> , <code>app.js</code>), environment configuration (<code>.env</code>), and MongoDB Mongoose database connection initialization (<code>db.js</code>).
Stage 2: Auth & Security	Implementation of User Schema, password encryption using <code>bcrypt.js</code> , token generation with <code>jsonwebtoken</code> , and Role-Based Access Control (RBAC) middleware guarding API routes.
Stage 3: Core Blog & CRUD API	Development of blog post routes, Mongoose models, and CRUD controllers allowing draft creation, publishing, reading, updating, and deleting blog entries along with category/tag queries.
Stage 4: AI Feature Integration	Integration of Google Gemini API SDK (<code>@google/generative-ai</code>), constructing <code>geminiService.js</code> , and mapping endpoints for AI blog post synthesis (<code>/api/ai/generate-blog</code>) and article summarization (<code>/api/ai/summarize</code>).
Stage 5: Verification & Testing	End-to-end endpoint verification via Postman / Thunder Client, error handling validation, environment security audits, and project documentation completion.

2. Team Responsibilities

- **Akash.M (Team Leader):** Overall project orchestration, core system architecture design, Express server setup (`server.js`, `app.js`), database connection (`db.js`), and integration testing.
- **Babu.K:** Authentication module implementation, User Mongoose Schema development, password hashing (`bcrypt.js`), and JWT authorization middleware.
- **Balasubramaniam.S:** Blog CRUD controller development, route definitions, category/tag indexing, and database query optimizations.
- **Benjamin Francis.C:** Google Gemini AI integration (`geminiService.js`), AI route implementation (`/api/ai/*`), and API client test collections.

3. Submission Plan

- **Milestone 1:** Architecture blueprint, folder initialization, and Mongoose database schema validation.
- **Milestone 2:** Operational Authentication system and protected Blog CRUD API routes verified via test suite.
- **Milestone 3:** Final submission of AI content tools, system testing, comprehensive phase documentation, and execution demonstration.